Industrial Component Anomaly Detection

Business & Strategy Report

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September 12, 2026

Abstract

In modern industrial manufacturing, automated visual quality assurance must resolve a fundamental operational conflict: balancing the downtime and scrap costs caused by false alarms against the severe financial liability of letting defective components reach customers. This report presents the commercial and technical case for an enterprise-grade, self-supervised Anomaly Detection Software-as-a-Service (SaaS) platform designed for discrete parts and continuous surface materials. Within this architecture, the inspection threshold is treated not as an arbitrary benchmark number, but as a practical business setting calibrated directly against each customer's actual costs of false alarms versus missed defects.

Benchmarked across all 15 categories of the MVTec Anomaly Detection dataset under our strict zero-test-leakage protocol (strictly reserving an 85% fitting partition and a 15% normal validation partition), our reference PatchCore engine with a frozen ResNet-18 backbone achieves a macro-average Image F_1-score of 0.929, PR-AUC of 0.988, and AUPIMO of 0.603. This decisively outperforms generative convolutional autoencoders while requiring no client-side backpropagation. New product parts can be onboarded through a single pass over standard reference images, and gradual factory changes (such as lighting shifts or tool wear) can be handled by updating the local reference set without retraining the frozen backbone.

Furthermore, the report introduces a clear economic cost model showing that high-speed commodity fasteners and life-critical pharmaceutical tablets require vastly different inspection thresholds based on their specific error costs. A three-tier calibration methodology translates these business risks into line-side sorting profiles. Deployment latency, hardware capacity, and payback must be validated for each plant using measured target-hardware throughput together with local labour, downtime, scrap, and integration costs; the MVTec benchmark alone does not establish a universal payback period.

1 Introduction

In modern high-throughput manufacturing, industrial quality assurance is persistently constrained by a fundamental operational conflict: balancing the direct scrap and downtime costs of false alarms against the downstream commercial and legal liability of uncontained defect escapes. Manual visual inspection remains limited by human fatigue, while conventional rule-based machine vision systems exhibit extreme parametric brittleness to supplier tolerances and ambient illumination shifts. To solve this, we present the technical and commercial justification for an enterprise-grade, self-supervised Anomaly Detection Software-as-a-Service (SaaS) platform. We position our architecture as an auditable, cost-sensitive risk-management pipeline. We evaluate components continuously against empirical anomaly score distributions, calibrating line-side divert thresholds directly against each customer's specific costs of false alarms versus missed defects.

We developed our technical foundation using the PatchCore architecture [4] with a frozen ResNet-18 neural network. Evaluated across the canonical MVTec Anomaly Detection benchmark [1, 2] under our strict zero-test-leakage protocol, our reference engine achieves a macro-average Image F_1-score of 0.929, PR-AUC of 0.988, and AUPIMO [6] of 0.603, decisively outperforming generative convolutional autoencoders. For plant directors, the relationship between these headline metrics warrants a plain explanation: the F_1-score is *dynamic* – measuring how profitably the system makes pass/reject decisions at a specific operational threshold – whilst AUPIMO is *fixed* – measuring how cleanly and reliably the model pinpoints defects independent of any threshold setting. In everyday terms: AUPIMO tells

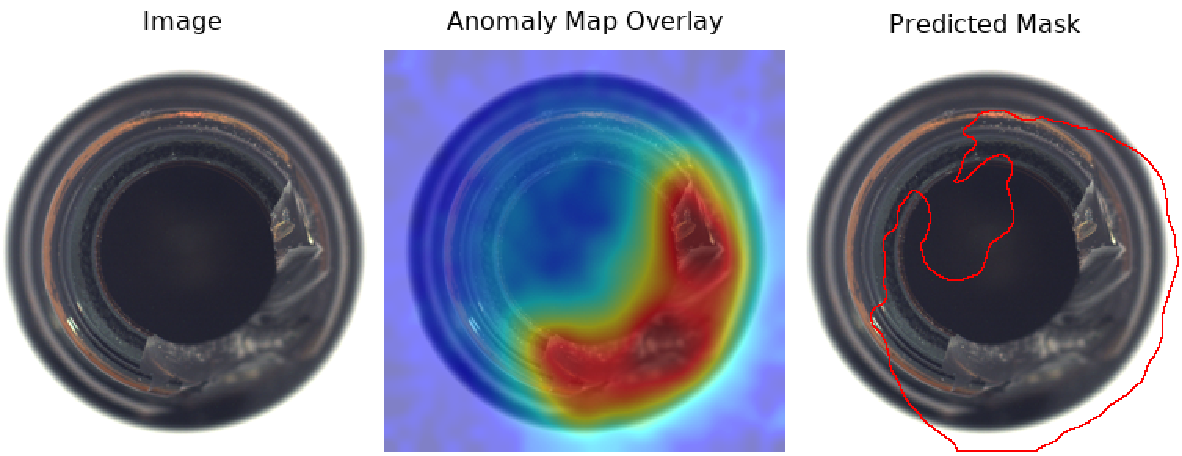

Figure 1: PatchCore explainability output for a representative defective bottle. From left to right: the input image, continuous anomaly-map overlay, and predicted contour mask at the frozen validation-calibrated operating threshold.

management how sharp the camera system's "eyes" inherently are, whilst the F_1-score measures how well that vision performs at your chosen factory setting.

The remainder of this report is structured as follows. Section 2 establishes the economic cost model underlying our threshold calibration. Section 3 characterises the MVTec AD benchmark and extracts the empirical data properties that govern metric selection. Sections 4 and 6 detail our model architecture and its empirical benchmark performance. Section 7 addresses hardware execution, explainability, and enterprise reliability. Section 8 concludes the report.

2 Error Asymmetry and Cost-Sensitive Decision Making

Quality assurance decisions in industrial series manufacturing cannot be evaluated using symmetric loss functions. Conventional evaluation paradigms in computer vision treat a missed non-conforming part (Type II error, defect escape) and a spurious rejection of a conforming part (Type I error, pseudo-scrap) as equivalent errors. In production engineering, this abstraction is fundamentally flawed. We assert that Type I and Type II errors induce radically disparate commercial, legal, and operational consequences. Disregarding this asymmetry results in a miscalibrated deployment that either induces severe productivity losses through excessive false alarms or exposes the enterprise to catastrophic downstream liability through uncontained defect escapes.

2.1 The Economic Cost Model

When setting inspection thresholds, we must balance the operational friction of false alarms against the severe liability of letting defective components reach downstream assembly or end customers. The economically rational objective is simple: we must choose the operating threshold that minimises total overall cost.

Let the expected total cost of quality at a given operational decision threshold t be denoted $C(t)$. Decomposing the loss across both error classes yields

$$C(t) = \mathrm{FP}(t) \cdot c_{\mathrm{FP}} + \mathrm{FN}(t) \cdot c_{\mathrm{FN}}, \tag{1}$$

where $\mathrm{FP}(t)$ and $\mathrm{FN}(t)$ denote the cumulative frequencies of false positives (pseudo-scrap) and false negatives (defect escapes) at threshold t, whilst c_{FP} and c_{FN} represent their respective marginal unit costs. We emphasise that the complexity of these involved factors is paramount: c_{FP} incorporates the downtime cost of halting a feed line, secondary labour for manual re-inspection, and the material loss of scrapping an in-spec component. Conversely, c_{FN} reflects the punitive exposure of an escaped defect: downstream tooling damage, assembly line stoppages, warranty replacements, regulatory penalties, and corporate brand damage.

Under a continuous and differentiable score distribution, we define the optimal threshold t^* that minimises total financial loss as:

$$t^* = \frac{c_{\mathrm{FP}}}{c_{\mathrm{FP}} + c_{\mathrm{FN}}}. \tag{2}$$

Equation 2 formalises the mathematical foundation of our threshold calibration engine. We do not assume fixed cost numbers; instead, we rely on financial parameters provided directly by the plant's controlling and quality management teams. We transform the inspection threshold from a rigid machine learning setting into a configurable, complex business decision.

2.2 The Industrial Liability Spectrum: Screws and Pills as Boundary Cases

We define the risk ratio $\rho = c_{\mathrm{FN}}/c_{\mathrm{FP}}$ to dictate the appropriate operational regime. Our systematic survey of industrial manufacturing sectors reveals that this ratio spans several orders of magnitude across active production lines, as illustrated in Table 1. Note that these cost profiles and risk ratios are illustrative estimates derived from composite industry case studies, intended to demonstrate the spectrum of liability rather than assert precise financial figures for specific manufacturers.

In high-volume commodity regimes (e.g., *Screw*), line stoppage costs dominate. If an isolated fastener with a minor blemish escapes into a bulk shipment, the loss is merely the low unit value of a single screw ($c_{\mathrm{FP}} \gg c_{\mathrm{FN}}$). We calibrate this gate with an $F_{0.5}$ objective, pushing the threshold towards unity to prevent excessive pseudo-scrap.

Conversely, in the life sciences sector (e.g., *Pill*), an undetected contaminated tablet triggers mandatory batch recalls and severe liability. Against this catastrophic exposure, the scrap cost of discarding conforming tablets is negligible. We calibrate this decision gate with an F_2 objective, lowering the threshold aggressively until full defect containment is achieved. We absorb a controlled false alarm rate as a calculated operating expense. By decoupling feature extraction from threshold calibration, we expose the operational decision gate as a dynamic, tunable financial parameter.

3 Industrial Data: Lessons from the MVTec AD Benchmark

To validate the platform under authentic factory conditions, we conducted all algorithmic development and empirical benchmarking on the MVTec Anomaly Detection (MVTec AD) dataset. This benchmark comprises 15 distinct manufacturing categories acquired in active production settings, covering continuous textures (e.g., *Carpet*, *Leather*) and rigid structural components (e.g., *Screw*, *Pill*).

3.1 The Accuracy Paradox and Its Implications for Metric Selection

Our exploratory data analysis of the benchmark exposes the fatal vulnerability of conventional accuracy metrics. We proved an extreme spatial imbalance: across the defective images, the median anomalous region accounts for merely 1.54% of the total image surface. This creates the *accuracy paradox*: a completely useless system that flags every single pixel as conforming achieves over 98.5% accuracy whilst failing to catch a single defect.

Table 1: Economic error asymmetry across representative industrial manufacturing categories.

Category	**False Alarm Cost**	**Defect Escape Cost**	**Risk Ratio (ρ)**	**Optimal t^***
Fasteners (*Screw*)	High (Line stoppage)	Low (Scrapped unit)	$\approx 4 \times 10^{-5}$	≈ 0.9999
Packaging (*Bottle*)	Low (Recycling)	High (Leakage cleanup)	$\approx 3 \times 10^{3}$	≈ 0.0003
Automotive (*Metal Nut*)	Low (Rework bin)	Very High (Recall)	$\approx 2.5 \times 10^{5}$	$\approx 4 \times 10^{-6}$
Life Sciences (*Pill*)	Very Low (Discard)	Catastrophic (Recall)	$\approx 10^{8}$	$\approx 10^{-8}$

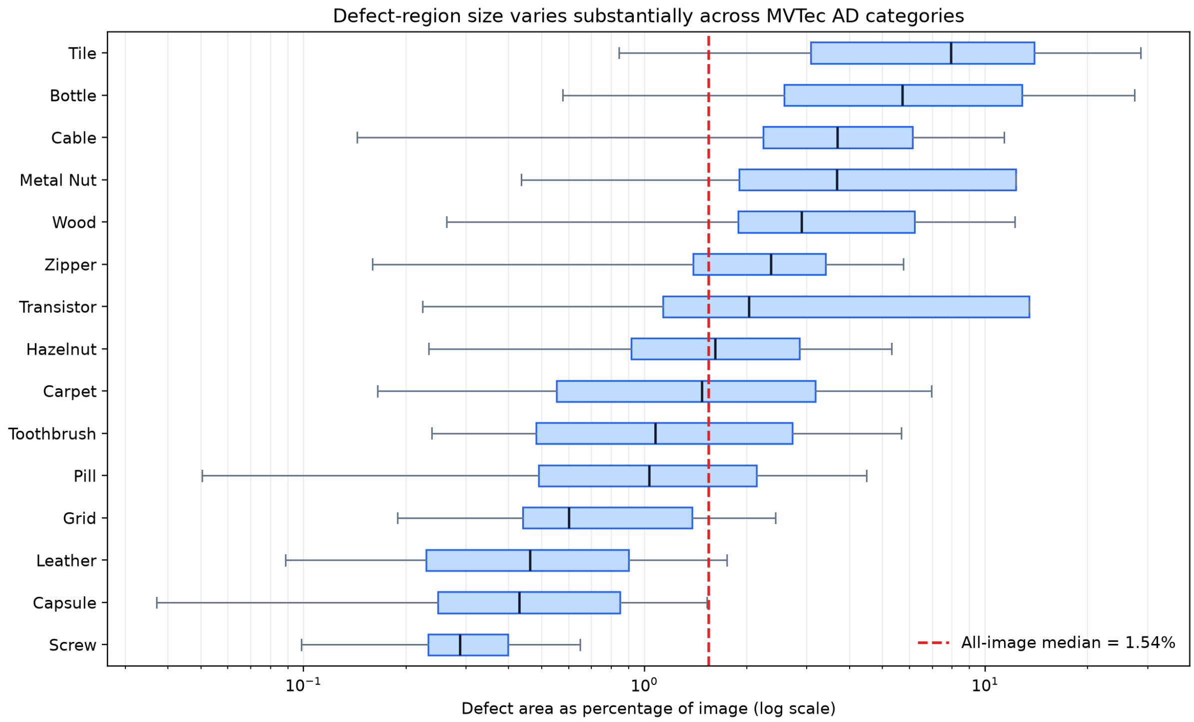

Figure 2: Per-image defect-area distributions computed from the 1,258 defective MVTec AD test masks. Boxes show category-level interquartile ranges on a logarithmic axis; the dashed line marks the all-image median of 1.54%.

Furthermore, we conducted a Kruskal-Wallis statistical test across the 15 categories, yielding an H-statistic of 626.5 ($p < 0.001$). This decisively rejects the hypothesis that defect sizes are uniform across products. Because defect scales vary by a factor of 43 across products, we established that no single universal threshold can reliably serve diverse production lines. The identical mathematical distortion invalidates standard pixel-level AUROC. For this reason, we evaluate our models using metrics that focus strictly on where defects actually reside: Image-level Precision-Recall Area Under Curve (PR-AUC) and the Area Under the Per-Image Overlap curve (AUPIMO) [6] over the ultra-low false-positive rate window $\text{FPR} \in [10^{-5}, 10^{-4}]$.

3.2 Heterogeneous Preprocessing and the Cold-Start Dilemma

Our exploration also revealed that industrial images are subject to measurable environmental drift, such as ambient lighting changes and conveyor vibration. If left uncorrected, an inspection system risks treating these normal lighting drifts as defects. We resolved this by tailoring data preprocessing to the physical characteristics of each product: employing affine transformations for continuous textures, and prioritising photometric normalisation with calibrated luminance jitter for rigid components.

Finally, we observed that supervised deep learning architectures are economically unfeasible in factory settings because defective components occur sporadically and unpredictably. We eliminated this "cold-start" barrier through self-supervised representation learning. By fitting the probability distribution over our categories exclusively on standard production output, we established an empirical baseline of the conforming state. We empower plant directors to commission a newly introduced production line on day one using only standard production parts, completely bypassing the costly defect collection trap.

4 Solution Architecture and Model Selection

We developed and empirically evaluated two architectural paradigms under identical experimental conditions across all 15 MVTec AD categories: the non-parametric PatchCore feature-matching framework and a deep generative convolutional autoencoder (the Keras CAE). We trained and calibrated both pipelines under the deterministic 85% fitting / 15% normal validation split with seed 42, evaluated at the canonical 256×256 spatial resolution, and scored against unseen evaluation splits under our strict zero-test-leakage protocol (described in Section 5).

4.1 PatchCore: Deep Feature Transfer and Coreset Memory Banks

We recommend PatchCore, utilising a frozen ResNet-18 backbone pre-trained on ImageNet, as the production-grade engine for plant-wide deployment. Beyond superior empirical detection metrics, we selected this architecture because it exhibits three decisive practical advantages that resolve longstanding operational pain points in manufacturing IT.

First, we eliminated client-side backpropagation. The core neural network (ResNet-18) acts as a fixed feature extractor. Once deployed, onboarding a new product type requires a single pass over its available standard reference images to build the memory bank. The elapsed onboarding time depends on reference-set size, target hardware, and coreset configuration and must therefore be measured during commissioning.

Second, our memory bank guarantees complete client data sovereignty and protection of intellectual property. We store conforming patterns locally in a reference memory bank rather than in network weights. Because this representation does not alter the underlying model, proprietary part geometries, CAD toolpaths, and confidential material formulations remain strictly confined to the local industrial personal computer (IPC). Furthermore, we employ a coreset selection algorithm to compress the stored reference library by an order of magnitude, guaranteeing fast inspection times that easily match conveyor speeds.

Third, we selected a comparatively lightweight backbone. Wide-ResNet-50-2 repeatedly exhausted the available GPU memory during the automated Optuna sweeps, so we standardised on ResNet-18 to preserve stable, reproducible execution. The final study did not perform a controlled backbone-accuracy comparison or a production IPC latency benchmark; memory capacity and cycle-time compliance must therefore be verified on the customer's target hardware before deployment.

4.2 Comparative Analysis: The Failure Modes of Generative Autoencoders

The alternative paradigm we evaluated is a fully convolutional autoencoder implemented in Keras, trained to reconstruct nominal images via a composite loss function combining the Structural Similarity Index Measure (SSIM) [5] and Mean Squared Error (MSE), augmented by masked image modelling (MIM). During inference, we use the pixel-wise reconstruction residual as the continuous anomaly map, from which we extract image-level classification scores via spatial aggregation.

Whilst generative reconstruction models offer conceptual appeal, we discovered they suffer from a fatal structural flaw in industrial practice. Pixel-level reconstruction objectives inherently minimise global image variance rather than local defect contrast. On high-frequency stochastic textures – such as *Wood* grain, *Carpet* pile, or *Leather* surfaces – the generative decoder yields slightly smoothed reconstructions of high-frequency spatial patterns. This slight blur generates elevated residual errors across conforming surfaces, inducing severe pseudo-scrap and artificially deflating the operational F_1-score. Table 2 quantifies this performance disparity across all 15 benchmark categories.

Table 2: Macro-averaged performance across all 15 MVTec AD categories under our strict zero-test-leakage protocol.

Architecture	Image F_1 ↑	Image PR-AUC ↑	AUPIMO ↑	Pixel AUROC ↑	Client Training
PatchCore (ResNet-18)	**0.929**	**0.988**	**0.603**	0.968	Memory-bank construction only
Keras CAE (MIM + SSIM+MSE)	0.495	0.867	0.058	0.875	Iterative GPU training

Empirically, the Keras CAE trails our PatchCore architecture by an order of magnitude on pixel-level AUPIMO (0.058 versus 0.603) and by 0.434 points in image F_1-score (0.495 versus 0.929). We confirmed this deficit cannot be mitigated through hyperparameter optimisation or alternative loss weighting; it reflects an intrinsic limitation of pixel-space generative modelling for industrial surface inspection.

4.3 Continuous Calibration: Mitigate Operational Drift Without Retraining

We designed the decoupled coreset architecture to absorb manufacturing distribution drift without requiring model retraining. When minor physical alterations occur on the line – such as a new steel batch exhibiting marginal surface roughness variations or illumination drift caused by lamp aging – conventional deep learning pipelines begin generating elevated anomaly scores for nominal parts. In a standard convolutional network, resolving this issue requires collecting fresh image batches, executing full backpropagation, and redeploying weights.

Within our PatchCore framework, this drift can be addressed at line side without retraining the frozen feature extractor. Quality assurance operators examine flagged units via the local inspection terminal and confirm eligible examples as nominal. After controlled review, the system can rebuild the coreset and recalibrate the decision threshold against a refreshed normal validation pool. Update duration and acceptance criteria are deployment measurements rather than outcomes of the final MVTec benchmark.

5 Operational Threshold Calibration

A pervasive failure mode in industrial computer vision deployments is the uncalibrated transfer of decision thresholds derived during laboratory model development directly onto the factory floor. In academic literature, algorithms are benchmarked using integrated area-under-the-curve metrics. Whilst indispensable for offline algorithm selection, these metrics provide zero guidance for operational commissioning. An industrial quality gate does not execute a continuous characteristic curve; it executes an instantaneous binary divert decision. We must calibrate the numerical threshold governing this decision to minimise total economic loss rather than to maximise an academic abstraction.

5.1 The Operational Distinction Between AUPIMO and Decision Thresholds

As outlined in Section 1, AUPIMO serves as a fixed benchmark of inherent optical sharpness, completely independent of where the pass/divert threshold is set. However, AUPIMO itself does not tell a conveyor mechanism when to push a part into a reject bin. We must calibrate an operational decision threshold to the factory's specific economic balance. A vision engine with outstanding inherent optical sharpness (AUPIMO = 0.603) will fail commercially if coupled to an arbitrary, uncalibrated threshold, whereas a model with an AUPIMO of 0.500 calibrated precisely to the customer's actual costs will deliver immediate operational value.

By the same token, we categorically discard pixel-level AUROC as an operational metric. Because conforming pixels make up roughly 98.5% of total component surface area, AUROC produces deceptively high scores regardless of true defect sensitivity. While reported in Table 2 to ensure benchmark reproducibility, it exerts zero influence on our operational threshold calibration.

5.2 Strict Zero-Test-Leakage Calibration Protocol

We mandate an immutable, audit-compliant calibration protocol that enforces strict isolation between model development, threshold derivation, and final acceptance testing. For every component category, we deterministically partition (seed 42) all available nominal training imagery into an 85% fitting partition, dedicated to coreset memory bank construction, and a 15% normal validation partition, held out exclusively for statistical threshold calibration. Crucially, the official evaluation split (the test set) remains completely untouched and unseen throughout the entire fitting and calibration lifecycle.

We derive the operational decision threshold τ empirically from the score distribution generated across the held-out normal validation partition. Formally,

$$\tau_\alpha = \mathrm{Quantile}_{1-\alpha}(\{s_i \mid x_i \in \mathcal{D}_{\mathrm{val,normal}}\}), \tag{3}$$

where x_i represents an individual image within the held-out normal validation set $\mathcal{D}_{\mathrm{val,normal}}$, s_i denotes the corresponding continuous anomaly score computed by the model for that image, and α denotes the plant's tolerable false-alarm budget (the Type I error ceiling). The final results in Table 3 use the category-specific 95th percentile of normal-validation image scores, corresponding to $\alpha = 0.05$; this threshold was frozen before official test inference.

Equation 2 and Equation 3 operate on different scales. The former is a probability decision rule under calibrated posterior probabilities, whereas the latter is a quantile of model-specific patch-distance scores. We therefore do not equate α with t^*. Instead, the cost ratio informs the choice among predeclared false-alarm budgets, and each selected budget is converted into a physical score boundary using only the normal validation distribution.

5.3 Standardised Risk-Tiered Operating Profiles

To operationalise this framework across diverse manufacturing sectors, we standardise three predefined risk tiers:

- **Tier 1: Life-Safety and High-Liability (e.g., Pharmaceuticals):** Governed by an F_2-optimised calibration objective ($\alpha \approx 0.10$). We weight defect recall four times more heavily than precision, shifting the decision threshold into the nominal distribution tail. We divert every borderline component to manual audit.
- **Tier 2: Precision Industrial (e.g., Automotive Powertrain):** Governed by a balanced F_1-optimised objective ($\alpha \approx 0.05$). We establish a cost-neutral equilibrium between Type I and Type II errors, appropriate for mid-tier risk environments.
- **Tier 3: High-Throughput Commodity (e.g., Threaded Fasteners):** Governed by an $F_{0.5}$-optimised objective ($\alpha \approx 0.01$). We weight precision four times more heavily than defect recall, preventing pseudo-scrap from choking downstream automated packaging equipment while catching severe structural anomalies.

Figure 3 illustrates how the three profiles select different quantiles of the same normal-validation distribution. The curves are schematic because the final report retains the frozen thresholds and metrics, but not the underlying per-image validation-score arrays.

6 Empirical Performance and Benchmark Results

Under our strict zero-test-leakage protocol, we executed exhaustive empirical evaluations across all 15 MVTec AD categories, calibrating operational decision thresholds exclusively on held-out normal validation partitions. The results documented below characterise our production-grade reference architecture: PatchCore utilising a frozen ResNet-18 backbone.

6.1 Category-Level Performance Breakdown

Table 3 provides the disaggregated performance across all 15 industrial categories. Evaluated at the frozen category-specific 95th-percentile normal-validation threshold ($\alpha = 0.05$), we achieved a macro-average Image F_1-score of 0.929, an Image PR-AUC of 0.988, and a pixel-level AUPIMO of 0.603. The unrounded category values average to defect recall 0.932 and precision 0.936.

To contextualise this performance, we compare each category with its own trivial 'all-defective' rule, whose Image F_1 is $2p/(1+p)$ for that category's defective prevalence p. PatchCore exceeds this category-specific reference in 13 of the 15 categories. The exceptions are *Pill* (0.891 versus a trivial 0.916) and *Screw* (0.796 versus 0.853). Their Image PR-AUC values remain high at 0.982 and 0.983, respectively, indicating strong score ranking despite weaker performance at the frozen validation-calibrated operating point.

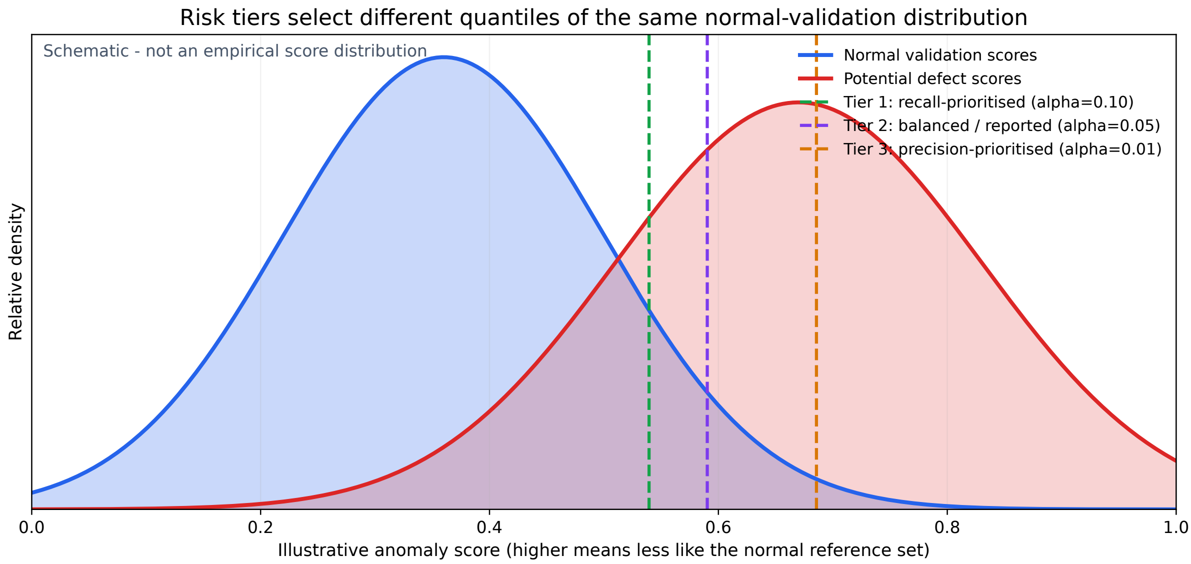

Figure 3: Schematic relationship between the normal-validation score distribution and the three predeclared operating profiles. The reported benchmark uses the Tier 2 setting, $\alpha = 0.05$; no empirical result is inferred from the illustrative curve shapes.

Table 3: PatchCore (ResNet-18) empirical performance across all 15 MVTec AD categories under our strict zero-test-leakage protocol. Image decisions use the category-specific 95th percentile of scores from the 15% normal validation partition ($\alpha = 0.05$), frozen before official test inference.

Category	**Image F_1**	**Recall**	**Precision**	**PR-AUC**	**AUPIMO**
Discrete Structural Objects					
Bottle	0.962	1.000	0.926	1.000	0.982
Cable	0.933	0.913	0.955	0.985	0.511
Capsule	0.982	0.982	0.982	0.996	0.563
Hazelnut	0.971	0.943	1.000	0.999	0.863
Metal Nut	0.963	0.989	0.939	0.998	0.760
Pill	0.891	0.816	0.983	0.982	0.293
Screw	0.796	0.672	0.976	0.983	0.380
Toothbrush	0.923	1.000	0.857	0.949	0.414
Transistor	0.916	0.950	0.884	0.976	0.427
Zipper	0.962	0.958	0.966	0.985	0.251
Continuous Textures					
Carpet	0.926	0.989	0.871	0.994	0.796
Grid	0.916	0.860	0.980	0.986	0.506
Leather	0.898	1.000	0.814	0.998	0.972
Tile	0.951	0.917	0.987	0.993	0.685
Wood	0.952	0.983	0.922	0.995	0.642
Macro Average	**0.929**	**0.932**	**0.936**	**0.988**	**0.603**

6.2 Engineering Analysis of Boundary Categories

The two categories presenting the most pronounced economic asymmetry – pharmaceutical tablets (*Pill*) and threaded fasteners (*Screw*) – warrant detailed scrutiny.

At the reported $\alpha = 0.05$ operating point, the *Pill* category attains an Image F_1-score of 0.891, recall of 0.816, and precision of 0.983. In a life-critical pharmaceutical facility, an 18.4% defect escape rate would be intolerable. Tier 1 therefore proposes a lower, recall-prioritised threshold and manual review of borderline cases. This is a commissioning policy to be validated on plant-specific acceptance data, not a separately benchmarked claim of perfect recall in Table 3.

Conversely, the *Screw* category is the most difficult image-level classification case, recording an Image F_1-score of 0.796 and AUPIMO of 0.380. Screws are photographed at arbitrary angles, and microscopic thread defects blend into normal machining marks. Against the final Keras CAE baseline, PatchCore raises Screw Image F_1 from 0.421 to 0.796 and AUPIMO from 0.005 to 0.380. At the reported operating point, precision is 0.976 and recall is 0.672; a distinct Tier 3 operating point would require separate commissioning validation.

6.3 Spatial Localisation Quality and Material Characteristics

While continuous texture categories like *Leather* (0.972) and *Carpet* (0.796) demonstrate strong spatial localisation fidelity under the AUPIMO metric, the highest score belongs to the discrete *Bottle* category (0.982); *Hazelnut* (0.863) also surpasses most texture classes. Localisation difficulty is not identical to image-level classification difficulty: *Zipper* has the lowest AUPIMO (0.251), followed by *Pill* (0.293) and *Screw* (0.380), whereas Screw has the lowest Image F_1. The corresponding Image PR-AUC values remain 0.985, 0.982, and 0.983, showing that category ranking can remain strong even when spatial boundaries or a frozen operating point are challenging.

7 Production Deployment: Hardware, Latency, Explainability

Deploying deep learning systems into active high-speed manufacturing environments introduces deterministic physical constraints entirely absent from offline academic research. For capacity planning, a representative cadence of five to thirty components per second implies a nominal latency budget of approximately 30–200 ms per component. These figures are illustrative engineering assumptions, not measurements from the MVTec evaluation. Each deployment must establish its actual takt time, end-to-end latency budget, and hardware acceptance criteria on the target line.

7.1 Hardware-Tiered Inference and Cycle-Time Profiling

The software can select execution settings according to locally available compute resources. The final scientific evaluation establishes detection quality, but it does not constitute a controlled CPU/GPU hardware benchmark.

By standardising on ResNet-18 after Wide-ResNet-50-2 exhausted the development GPU memory during Optuna sweeps, we reduced deployment risk relative to the heavier candidate. Whether feature extraction and nearest-neighbour search meet a plant's throughput target on a fanless Industrial Personal Computer (IPC) remains an acceptance-test question. A retrofit may avoid a dedicated GPU where measured CPU throughput satisfies that line's takt time; otherwise GPU acceleration remains an available deployment option.

Where edge GPU acceleration is present, the same pipeline can use batched execution, subject to target-hardware profiling and end-to-end line validation.

7.2 Explainable AI (XAI) and Operator Interface Architecture

An uninterpretable divert signal introduces a severe operational vulnerability: operator override. If line-side quality inspectors cannot discern the physical rationale behind an automated rejection, they routinely dismiss the alert. Over prolonged operations, this breeds chronic skepticism and degrades inspection coverage back to the unreliable manual baseline.

We counteract this vulnerability by providing continuous, pixel-level spatial explainability. Utilising the nearest-neighbour patch distance tensor, our runtime renders a calibrated, colour-coded anomaly heatmap that pinpoints the exact spatial coordinates driving the elevated score. Applying the calibrated decision threshold τ, we derive a binarised defect contour mask, providing the line operator with both an unambiguous anomaly score and an auditable spatial justification. We rescale both outputs to the canonical 256×256 resolution prior to display, guaranteeing precise geometric alignment with physical part coordinates. Figure 4 illustrates representative defect heatmaps and contour overlays.

We deploy the line-side interface as a reactive Streamlit application. Quality engineers can inspect diverted components in real time, examine the spatial heatmap justification, and validate or reclassify the divert. This feedback action appends the verified embeddings to the local memory bank, transforming

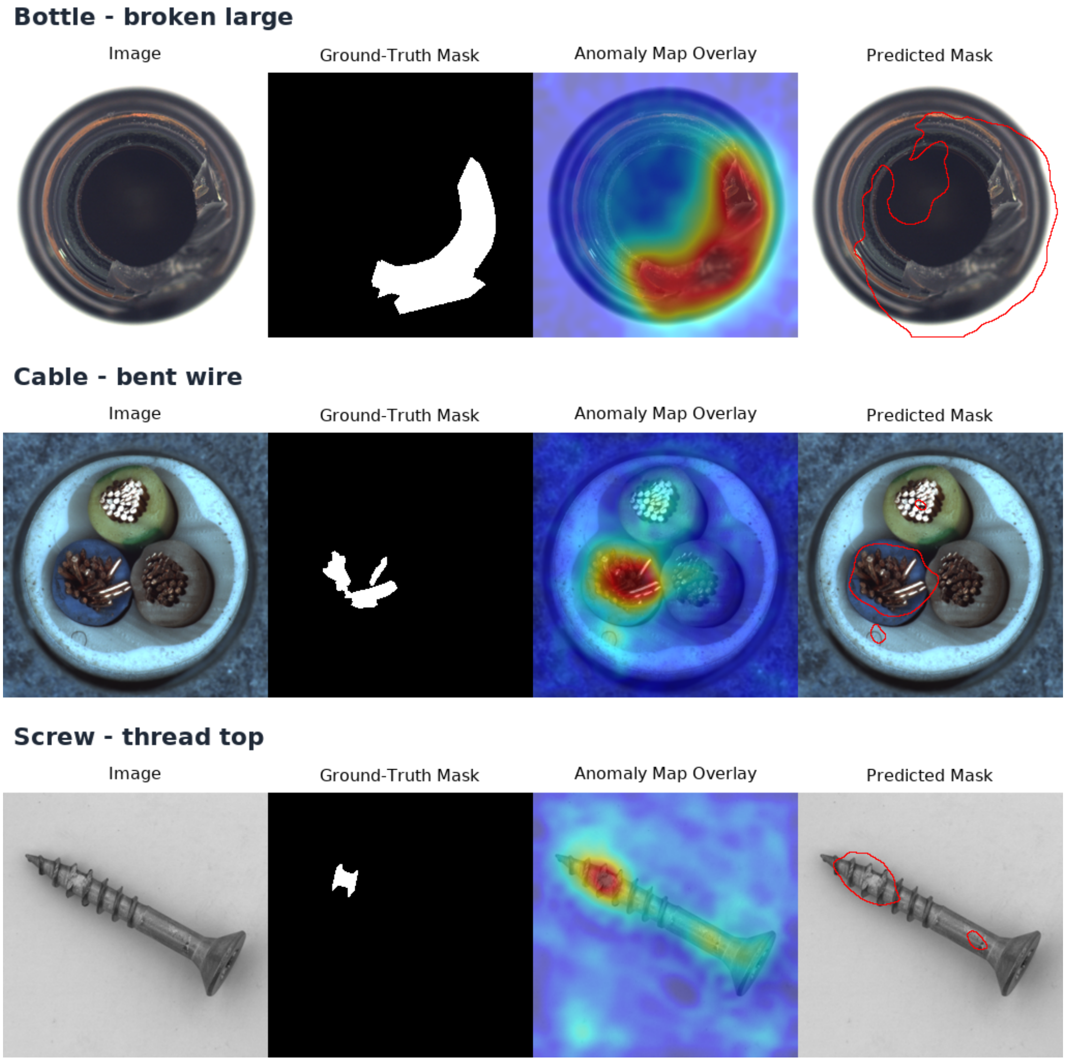

Figure 4: Stored PatchCore qualitative outputs for representative Bottle, Cable, and Screw defects. Each row shows the input, ground-truth mask, anomaly-map overlay, and predicted contour mask.

line operators into active stewards of continuous system refinement whilst requiring zero machine learning background.

7.3 Enterprise Software Architecture and Intellectual Property Safeguards

Beyond inference latency, we designed the platform to satisfy rigorous IT governance, reliability, and cybersecurity criteria. We built the platform upon a strictly reproducible software supply chain. We deterministically resolve environment dependencies using `Pixi`, and orchestrate operational tasks via `Just`. This guarantees that software installations execute deterministically across distributed edge nodes without library collisions.

Furthermore, we adhere to strict corporate IP and legal compliance standards. We constructed the core runtime exclusively upon permissively licensed, enterprise-compliant open-source frameworks (notably PyTorch and Anomalib [10] under Apache 2.0 and MIT licences). We strictly exclude viral copyleft licences (such as GPL/AGPL) that could legally threaten client intellectual property. This guarantees legal security for corporate procurement and ensures that manufacturing designs remain strictly confidential.

8 Conclusion

In this report, we established a robust anomaly detection platform for industrial computer vision, anchored by the cost-derived threshold system. By mathematically unifying our economic cost model with empirical validation metrics, we demonstrated that the inspection threshold is not an abstract statistical parameter, but a practical, risk-tiered business decision. We calibrated this threshold directly against the real financial consequences of false-positive scrap costs versus false-negative defect escapes on the production line. Because a stripped fastener thread and a contaminated pharmaceutical tablet differ in their defect escape liability by orders of magnitude, this cost-derived framework empowers plant managers with a single, transparent lever to balance operational risks safely and optimally.

We built the underlying engineering architecture to execute this operational vision efficiently. We deployed a PatchCore engine with a frozen ResNet-18 backbone, delivering strong macro-average performance across diverse industrial categories without requiring complex model training on factory computers. We decoupled the coreset memory bank so that raw material variations and ambient illumination drift can be addressed by refreshing the local reference set without retraining the frozen backbone, and we engineered high-resolution spatial explainability overlays to render every divert decision auditable by line-side quality inspectors.

Looking forward, we aim to extend this foundation to encompass multi-camera perspectives and continuous online learning, adapting to more complex multi-part assemblies without manual intervention. By systematically replacing arbitrary heuristics with transparent, economically grounded algorithms, we have successfully transformed automated visual quality assurance from an unpredictable engineering experiment into an auditable, deterministic, and highly reliable operational asset for the manufacturing industry.

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A Post-Project Extension: Self-Supervised Vision Transformer Baselines

Explanatory Note: These architectures were evaluated iteratively after the primary project evaluation and therefore carry the same test-informed caveat as earlier prototypes. They pre-date the definitive freezing of the strict zero-test-leakage protocol and are included only for technical completeness and future direction; they did not form part of the benchmarked production scope in the core report.

A.1 Architectural Formulation and Literature Context

We evaluated two frozen self-supervised Vision Transformer (ViT) baselines to explore representation ceilings beyond convolutional features. For literature context, *Dinomaly* by Guo et al. [8] reports 99.6% image-level AUROC and 98.2% pixel-level AUROC on MVTec AD using foundation-model features in a purpose-built reconstruction architecture.

A.1.1 Self-Supervised Baselines (ViT-S/14)

We extracted patch-level token representations from the final encoder block of two off-the-shelf Vision Transformer models:

- **DINOv2 (ViT-S/14, frozen):** Patch-level cosine nearest-neighbour matching against an empirical memory bank of frozen self-supervised ViT patch tokens, pre-trained via discriminative self-distillation on the curated LVD-142M dataset [7].
- **DINOv3 (ViT-S/14, frozen):** An identical pipeline substituting DINOv3 weights [9], which introduce scaled pre-training and Gram anchoring for dense-feature quality.

Both use PatchCore's non-parametric memory-bank paradigm – zero client-side backpropagation and single-pass onboarding – while replacing the convolutional backbone with a self-attention transformer encoder; neither introduces the specialised reconstruction architecture used by *Dinomaly*.

A.2 Indicative Empirical Results and Engineering Trade-Offs

The exploratory outputs were DINOv2: Image $F_1 = 0.941$, PR-AUC = 0.989, AUPIMO = 0.602, and pixel AUROC = 0.969; and DINOv3: 0.957, 0.993, 0.680, and 0.967, respectively. For context only, DINOv2 pixel AUROC is 1.3 percentage points below Dinomaly's published 98.2%; this is not a protocol-matched ranking because the architecture and evaluation protocol differ.

These test-informed DINO runs were not subject to the clean, single-pass protocol lock used for the production benchmark. Nor did the project complete a controlled DINO-versus-ResNet memory and latency benchmark on a common industrial PC, so no quantitative model or hardware ranking is claimed.

Vision Transformers remain a promising research direction, particularly in purpose-built architectures such as Dinomaly. The frozen ResNet-18 PatchCore engine remains the deployment reference because it was evaluated under the clean final protocol; ViT deployment claims require a controlled benchmark on target industrial PC hardware and live drift conditions.

Table 4: Macro-averaged results from the exploratory DINOv2 and DINOv3 runs. Because these runs were test-informed, the values are reported for transparency and must not be ranked against the clean-protocol PatchCore reference.

Architecture	Image F_1 ↑	Image PR-AUC ↑	AUPIMO ↑	Pixel AUROC ↑
DINOv2 (ViT-S/14, frozen)	0.941	0.989	0.602	0.969
DINOv3 (ViT-S/14, frozen)	0.957	0.993	0.680	0.967